A Comparative Study of Deep Learning Architectures for Face Recognition

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Abstract

Face recognition is a significant area of computer vision that has various practical applications, including security systems, smartphone unlocking, and photo tagging. Over the past few years, advances in deep learning have considerably improved the performance of face recognition systems. This report examines four widely recognized deep learning approaches in this field: DeepFace, FaceNet, VGG Face, and ArcFace. DeepFace combines 3D facial alignment with a deep neural network trained to identify thousands of individuals. FaceNet generates a compact, 128-dimensional representation of a face by employing a triplet loss function. VGG Face is associated with a large facial image dataset and a deep convolutional neural network trained using that data. ArcFace introduces an additive angular margin loss function that helps produce more discriminative facial features. For each approach, the report discusses its fundamental concept, network architecture, training methodology, and performance on established benchmarks, including LFW and YTF. Finally, the four methods are compared to highlight their respective advantages and limitations.

I. Introduction

Face recognition is a computer vision technique used to identify or verify individuals from facial images or videos. It is widely used in biometric applications, including smartphone unlocking, airport security, attendance systems, social media tagging and video surveillance. It involves two main tasks: face verification, which checks whether two face images belong to the same person (one-to-one matching), and face identification, which determines a person's identity by comparing their face against a database of known individuals (one-to-many matching).

Although humans recognize faces easily, this task remains challenging for computers. The same person may look very different across images due to changes in pose, lighting, facial expression and age. Occlusions caused by glasses, masks or hair can further affect facial appearance. Meanwhile, different individuals may look similar. Consequently, early systems performed well under controlled conditions, such as frontal views and good lighting, but struggled with images captured “in the wild”.

Face recognition has been researched for decades. Early methods represented faces using a small set of numerical features learned from training images, while later approaches used hand-crafted features, such as local texture patterns, with classifiers like Support Vector Machines (SVMs). A major milestone was the release of the Labeled Faces in the Wild (LFW) dataset [1] in 2007. It contains over 13,000 face images of 5,749 people collected from news photographs and became a standard benchmark for face verification in unconstrained conditions. In 2011, the YouTube Faces (YTF) dataset [2] was introduced for video-based face verification. It contains 3,425 video clips of 1,595 people, often with blurry, low-quality frames. These datasets enabled fairer comparisons between different methods.

Around 2012, deep learning, particularly convolutional neural networks (CNNs), began outperforming hand-crafted features in many computer vision tasks. CNNs learn useful features directly from image pixels when trained on sufficient data, while GPUs made training large networks practical. Researchers soon applied CNNs to face recognition, rapidly improving accuracy on LFW.

This report examines four major deep learning approaches to face recognition. DeepFace [3], developed by Facebook in 2014,

combines 3D face alignment with a nine-layer neural network trained on approximately 4 million images, achieving 97.35% accuracy on LFW, close to human-level performance. FaceNet [4], introduced by Google in 2015, directly learns a 128-dimensional face embedding using triplet loss rather than training a classifier, achieving 99.63% accuracy on LFW. VGG Face [5], developed at the University of Oxford in 2015, addressed the limited availability of public face datasets by creating a dataset of 2.6 million images and training a deep CNN that achieved competitive results. ArcFace [6], proposed in 2019, introduced the additive angular margin loss function and achieved 99.83% accuracy on LFW.

Table 1 summarizes the four methods. DeepFace and VGG Face initially train their networks as classifiers using softmax loss, FaceNet employs triplet loss, and ArcFace uses softmax loss with an angular margin. Across these approaches, LFW accuracy increased from 97.35% to 99.83%. Sections II–V describe the four architectures, Section VI compares their performance and features, and Section VII presents the conclusion.

Table 1. Summary of the four architectures

Method (year)	Developed by	Training data	Loss function
DeepFace (2014)	Facebook	4.4M images, 4,030 people	Softmax
FaceNet (2015)	Google	100–200M images, ~8M people	Triplet loss
VGG Face (2015)	University of Oxford	2.6M images, 2,622 people	Softmax + triplet loss
ArcFace (2019)	Imperial College London	5.8M images, 85K people	Additive angular margin loss

II. Architecture 1: DeepFace

DeepFace was introduced by Taigman et al. [3]. It follows the standard face recognition pipeline of detection, alignment, representation and classification, with improvements mainly in alignment and feature representation.

DeepFace performs alignment using a 3D facial model. Initially, six key points around the eyes, nose and mouth are detected for basic 2D alignment. Next, 67 facial landmarks are matched to a generic 3D face model, and the image is transformed to make the face appear front-facing. This process, known as frontalisation, produces a 152×152 RGB image.

The network consists of nine layers. Its first three layers, comprising two convolutional layers and one max-pooling layer, extract basic features such as edges and textures. These are followed by three locally connected layers, which learn separate filters for different image locations to capture variations between facial regions, such as the eyes and mouth. The final two layers are fully connected. The first fully connected layer produces a 4,096-dimensional face representation. Overall, the network contains more than 120 million parameters and is trained to classify 4,030 individuals using 4.4 million images, the softmax function and cross-entropy loss:

$$p_k = \frac{exp(o_k)}{\sum_h exp(o_h)}, \quad L = -log\, p_k \qquad (1)$$

Here, o_k represents the network output for class k, while k denotes the correct class. During verification, two face representations are compared using the inner product or a weighted χ^2 distance. DeepFace achieved 97.35% accuracy on LFW using an ensemble of networks and 91.4% on YTF. The authors also demonstrated the importance of alignment: without it, accuracy dropped to 87.9%.

III. Architecture 2: FaceNet

FaceNet was introduced by Schroff et al. [4] at Google. Earlier approaches, such as DeepFace, trained a classifier and used one of its hidden layers to represent each face. The authors considered this approach indirect and noted that it produced large feature representations. In contrast, FaceNet uses a deep CNN to directly map each face image x to a 128-dimensional vector $f(x)$, known as an embedding, with $\|f(x)\|_2 = 1$. The network is trained to minimize the squared distance between embeddings of the same person while maximizing the distance between those of different individuals. Face

verification is then performed by checking whether the distance between two embeddings falls below a specified threshold.

FaceNet uses triplet loss during training. Each triplet consists of an anchor image x_{ia}, a positive image x_{ip} of the same person, and a negative image x_{in} belonging to a different person. The loss function is:

$$L = \sum_{i=1}^{N} \left[d\left(x_i^a, x_i^p\right) - d(x_i^a, x_i^n) + \alpha \right]_+ \quad (2)$$

Here, $d(x, y) = \|f(x) - f(y)\|_2^2$, and α represents a margin set to 0.2. Selecting suitable triplets is important because most are too easy to provide useful learning signals. FaceNet uses "semi-hard" negative examples, which are farther from the anchor than the positive example but remain within the margin. These are selected from large mini-batches containing approximately 1,800 images.

The authors evaluated two network architectures: a 22-layer Zeiler–Fergus model containing 140 million parameters and a smaller Inception-based model with only 7.5 million parameters, which achieved similar accuracy. FaceNet was trained on 100–200 million facial images of approximately 8 million individuals, using tight face crops instead of complex alignment. It achieved 99.63% accuracy on LFW and 95.12% on YTF.

IV. Architecture 3: VGG Face

VGG Face was introduced by Parkhi et al. [5] at the University of Oxford. The authors observed that leading methods relied on large private datasets owned by companies such as Facebook and Google, limiting access for other researchers. Their main contribution was a method for creating a large face dataset with minimal manual effort. They collected a list of 5,000 celebrities from IMDb and downloaded images using Google and Bing image searches. Automatic filtering and manual inspection were then used to clean the images, producing a dataset of 2.6 million images representing 2,622 individuals.

The network uses the VGG architecture, consisting of 13 convolutional layers with 3×3 filters and three fully connected layers. It accepts face images of size 224×224 pixels. Initially, the network is trained to classify the 2,622 individuals using softmax loss. The final layer is then retrained using triplet loss, as described in Eq. (2), to produce a 1,024-dimensional embedding better suited for face verification.

The authors reported that adding triplet loss reduced the error on LFW by 68%. They also found that training on a larger, less thoroughly cleaned dataset produced better results than using a smaller, fully cleaned dataset. VGG Face achieved 98.95% accuracy on LFW and 97.3% on YTF, using considerably less data than FaceNet and a single network.

V. Architecture 4: ArcFace

ArcFace was introduced by Deng et al. [6]. The authors noted that softmax loss does not produce sufficiently discriminative features for face recognition, while triplet loss is difficult to train due to the large number of possible triplets. To address these limitations, they proposed a new loss function called Additive Angular Margin Loss.

ArcFace normalises both the feature vectors and the weights of the final layer, making each class output depend only on the angle θ_j between the feature vector and its corresponding class weight. The feature vectors are then scaled by a fixed value s, and an angular margin m is added to the angle of the correct class. This increases the difficulty of classification during training and encourages the network to learn more compact and discriminative features for each individual. The loss function is:

$$L = -\frac{1}{N} \sum_{i=1}^{N} log \frac{e^{s\cos\left(\theta_{y_i} + m\right)}}{e^{s\cos\left(\theta_{y_i} + m\right)} + \sum_{j \neq y_i} e^{s\cos\theta_j}} \quad (3)$$

where θ_{yi} is the angle for the correct class y_i, s = 64 and m = 0.5. The authors explain that this margin corresponds to a geodesic distance on a hypersphere, and they compare it with the margins used in SphereFace and CosFace.

ArcFace uses a ResNet-100 network that outputs a 512-dimensional feature, and the input faces are 112×112 pixels. It was trained on the MS1MV2 dataset, which is a cleaned version of MS-Celeb-1M with 5.8 million images of 85,000 people. ArcFace achieved 99.83% on LFW and 98.02% on YTF, and it also performed very well on harder datasets such as MegaFace and IJB-C.

VI. Analysis and Discussion

Table 2 compares the four methods. All of them use deep CNNs trained on large datasets, but they differ in several important ways.

Table 2. Comparison of the four methods

Method	Alignment	Embedding size	LFW (%)	YTF (%)
DeepFace	3D	4,096	97.35	91.4
FaceNet	Tight crop only	128	99.63	95.12
VGG Face	2D	1,024	98.95	97.3
ArcFace	2D (5 points)	512	99.83	98.02

Loss function: DeepFace uses softmax loss, FaceNet uses triplet loss, VGG Face combines both, and ArcFace applies softmax loss with an angular margin. These choices significantly affect accuracy. VGG Face reduced its error by adding triplet loss [5], while ArcFace outperformed triplet loss when both used identical training data and network architectures [6]. ArcFace is also easier to train because it eliminates triplet selection.

Alignment: DeepFace relies on 3D alignment [3], whereas later methods use simple 2D alignment or face cropping. This suggests that deeper networks and larger datasets reduce the need for complex alignment.

Data: FaceNet used up to 200 million images, while VGG Face achieved competitive results with 2.6 million images, and ArcFace achieved the highest accuracy with 5.8 million images. Thus, data quality and loss function selection matter as much as dataset size. DeepFace and FaceNet use private datasets, making their results difficult to reproduce.

Limitations: The methods were trained on different datasets, making comparisons imperfect. Furthermore, most methods now exceed 99% accuracy on LFW, highlighting the need for harder benchmarks to reveal meaningful performance differences.

VII. Conclusion

This report examined four major deep learning architectures for face recognition. DeepFace demonstrated that 3D alignment and deep networks can approach human-level performance. FaceNet introduced compact embeddings using triplet loss, while VGG Face achieved strong results with a public dataset and a deep network. ArcFace introduced additive angular margin loss and achieved the highest accuracy among the four methods. Overall, face recognition has progressed towards simpler alignment, improved loss functions and larger datasets. Future systems should also prioritize fairness, privacy and reliable performance under challenging real-world conditions.

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